

Sanithu Heengama

Personal Information

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Google Scholar https://scholar.google.com/citations?user=tO_ezy0AAAAJ&hl=en

Education

08/2023–05/2027 **Sc.B., Computer Engineering (Hon)**
GPA: 3.8 | Brown University, Providence, RI, USA

Awards & Honors

- 3 x Undergraduate Teaching and Research SPRINT Fellowship
- Hong Kong International Mathematical Olympiad (HKIMO) Global Final- Aug 2023- World Star and Bronze Medal
- 2 x International Scholars with Honors, 2× Pearson Edexcel Excellence Awards

Publications

* represents equal contribution.

- The Journal of Emerging Investigators (JEI)- JEI-23-024R4: A cost-effective IoT-based intelligent indoor air quality monitoring -. Sanithu Heengama* (Accepted and published on Sep 05, 2023). Link to article., DOI: <https://doi.org/10.59720/23-024>

Research and Project Experience

08/2025–Current | **3D from multi-view and sensors**

Brown University | Mentors: Prof. Benjamin Kimia, Chiang-Heng Chien (Ph.D. Student)

- Focused on the theoretical, algorithmic, and engineering foundations of reconstructing 3D structures from multiple 2D images
- Focus on reconstructing 3D scenes from multiple camera views, addressing mathematical and computational challenges
- Cover camera models, including intrinsic and extrinsic parameters.
- Perform camera calibration to relate 3D world coordinates to 2D image coordinates.
- Apply epipolar geometry and compute essential/fundamental matrices for relating multiple views.
- Estimate relative camera poses between views.
- Use triangulation methods to recover 3D point positions from corresponding 2D points.
- Employ optimization techniques to improve reconstruction accuracy and robustness.

07/ 2026– Current | **Utilizing 3D Slicer for Musculoskeletal Joint and Implant Tracking**

Biomedical Engineering and The Warren Alpert Medical School | Mentors: Prof. J.J. Trey Crisco III,

- Extended and validated open-source 3D Slicer tools (SlicerAutoscooperM, Hierarchical 3D Registration) for markerless *in vivo* skeletal kinematic tracking from biplane videoradiography and 3DCT/4DCT imaging
- Assessed tracking accuracy across multiple skeletal structures and joint implants using Bland-Altman statistical analysis benchmarked against marker-based methods
- Applied multi-view 3D reconstruction and camera calibration principles to biplane radiographic pose estimation workflows
- Collaborated with a multi-institutional research consortium (Brown, Mayo Clinic, Kitware) supporting global adoption of open-source musculoskeletal tracking software

09/ 2025–05/2026 | **Esophageal Manometry Redesign**

Brown University, RISD University | Mentors: Dr. Celinda Kofron (Academic Advisor), Dr. Wai-Kit Lo, M.D (Clinical Advisor)

Co-leading a year-long effort to convert Redesigning High Resolution Esophageal Manometry (HRM) into a patient-friendly, clinically robust device through new sensing/form factors, embedded electronics, and

clinician-centric software, validated against today's standard of care. Scope spans requirements capture, prototyping, UX/human-factors testing, manufacturability, hospital integration, and IP planning for a functioning medical device prototype.

05/ 2025 -12/2025 | **Radiance Fields for 3D Computer Vision and Artificial Intelligence**

Brown University | Mentors: Prof. Srinath Sridhar, Research Assistant Angela Xing

- Collect visuotactile dataset on single-hand and bimanual manipulation of deformable objects.
- 3D print, assemble UMI gripper and integrate tactile sensors into the Brown Interaction Capture System (BrICS).
- Implement deformable 3D gaussian splatting reconstruction for ground truth particle position and velocity.

Sept 2025 -- Jan 2026 | **Development of AI Physics Game using Reinforcement Machine Learning and Physics-Informed Neural Nets**

Brown University | Mentors: Prof. Gaitskill

- Refine black-hole orbital simulation defining states, actions, and rewards.
- Implement RL agents to maintain stable orbits avoiding the black hole and boundaries.
- Develop multi-agent competitive environment with gravitational interactions.
- Optimize RL algorithms for better robustness and efficiency.
- Explore advanced physics models and neural policy adaptations.

Teaching

01/2025 - 05/2025 **Undergraduate Teaching Assistant**

ENGN 40 Dynamics and Vibrations

08/2024 - 12/2024 **Undergraduate Teaching Assistant**

ENGN 32 Introduction to Engineering: Design

Presentations

1. Sanithu Heengama*, Muzala Kawatu*, and Ezra Seidel. *Reaction-Diffusion Pattern Formation theory, linking mathematical models to chemical/biological applications*. Presented at DRP 2025 Workshop

Training

07/ 2025 -08 2025 | **Laboratory for Physical Sciences (LPS) Qubit Collaboratory (LQC) Quantum Computing**

Successfully completed intensive summer program covering quantum circuits, algorithms, error correction, and qubit technologies certification, with lectures and interactive sessions led by leading quantum scientists.

Skills

Technical Skills

Expert: Python, C++, MATLAB, PyTorch, PyTorch Lightning, scikit-learn, NumPy, SciPy, Verilog,

- Fluent: Java, TypeScript, JavaScript, React, Fusion 360, SolidWorks, Golang Cadence, ML, RNL

Language proficiency in English,

Leadership & Extracurriculars

- **Brown Science Olympiad**

Brown Science Olympiad (BSO) is a Brown University undergraduate organization that hosts annual invitational tournaments for middle and high school Science Olympiad teams, featuring 23 STEM events in chemistry, physics, anatomy, engineering, and more. I have volunteered as a supervisor, helping ensure smooth event operations and supporting participants throughout the competitions

- **Brown Space Engineering (BSE)**

Brown Space Engineering (BSE) is a Brown University student organization designing, building, and launching open-source, low-cost satellites like CubeSats to democratize space access for all students, regardless of experience. Active member.

Interests: Chip Design, Electronics, Reading